

RUTIK KOLAWALE

Solapur, Maharashtra | +91 80100 95224 | rutik.kolawale8010@gmail.com
Linkedin: linkedin.com/in/rutik-kolawale | GitHub: github.com/RUTIKKOLAWALE

PROFESSIONAL SUMMARY

Aerospace engineering graduate specializing in UAV systems integration, with hands-on experience across ArduPilot/Cube Orange flight controllers, MAVLink, embedded electronics (ESP32, STM32), and Fusion 360 mechanical design. Has built and flight-tested multiple multirotor and fixed-wing platforms end-to-end, from CAD and wiring to firmware tuning. Seeking a UAV Integration Engineer role to apply this systems-level hardware and software experience.

TECHNICAL SKILLS

- Flight Controllers & Firmware: ArduPilot, Pixhawk, Cube Orange, Mission Planner, QGroundControl
- Protocols & Bus Systems: MAVLink, I2C, UART, ELRS, RTK GPS/GNSS
- Embedded Development: ESP32, STM32, PlatformIO, Raspberry Pi
- Software & Tools: C, C++, Python, Qt, Mission Planner
- Design Software: Autodesk Fusion 360, SolidWorks

WORK EXPERIENCE

Experiments in Flights — Flight Laboratory | Indian Institute of Technology, Kanpur

June 2024

- Calculated aircraft weight and center of gravity (CG) for flight performance analysis.
- Gained hands-on experience in flight mechanics and aircraft stability analysis through experimental testing.
- Assisted in conducting experimental testing and data analysis for flight experiments.

Intern | Paras Anti-Drone Technologies, Nerul, Navi Mumbai

February 2026-August 2026

- Executed UAV flight testing for validation of anti-drone systems.
- Conducted hardware validation and software testing of anti-drone products using proprietary tools.
- Worked in embedded systems: sensor calibration, hardware validation, and diagnostics.

KEY PROJECTS

Hawk Green & Eagle Red — Multirotor Builds

- Built two independent drone platforms, each configured with a distinct telemetry link, for comparative range and interference testing.
- Hawk Green: 433 MHz telemetry link tuned for longer-range, lower-bandwidth link margin. Eagle Red: 2.4 GHz telemetry link tuned for higher-bandwidth, shorter-range performance.

Responsive Web Application — HTML5, CSS3, JavaScript, Bootstrap

- Developed a structured multi-page website using HTML5, CSS3, Bootstrap, and JavaScript with responsive design principles.
- Implemented semantic HTML structure and mobile-first layout, improving accessibility and cross-device compatibility.

RC Fixed-Wing Aircraft

- Designed and fabricated a radio-controlled fixed-wing aircraft from scratch with a Depron foam airframe, applying structural and aerodynamic design principles.
- Implemented control surface layout based on stability and CG analysis; performed iterative flight testing and tuning for stable flight.
- Achieved stable controlled flight across multiple test sessions, validating aerodynamic design and control system implementation.

UAV Canopy Design — Autodesk Fusion 360

- Designed a 150 × 150 × 70 mm lightweight UAV canopy in Fusion 360 for an anti-drone platform, optimized for FDM 3D printing in PETG/ABS.
- Engineered mounting provisions including a Ø69 mm GPS pocket, safety switch housing, and trapezoidal dual-screw mounting tabs, maintaining a 2 mm wall thickness for optimal strength-to-weight ratio.
- Applied Design for Manufacturing (DFM) principles — fillets, chamfers, wall-thickness optimization — to minimize print supports and improve structural integrity.

EDUCATION

Bachelor of Technology in Aerospace Engineering

Ajeenkya DY Patil University, Pune, Maharashtra

2022 – 2026 | CGPA: 7.9

LANGUAGES

Hindi | English | Marathi | Gujarati